

Answer to the question no. 1

The primary distinction between traditional and agile approaches is the order of the project phases. The sequence of the phases in which the project is developed is linear in structured development methodologies, but iterative in agile.

The differences between structured and agile development is given below:

Structured development	Agile development
1] Usually it is used to develop simple software.	1] It is used to develop complex software.
2] It is based on linear organization structure.	2] It is based on iterative organizational structure.
3] It supports a fixed development model.	3] It supports a changeable development model.
4. It is consist of five phase.	4. It is consist of three phase.
5. Development cost is less because scale of the project is small or medium	5. Development cost is higher because the project is large.

Ans to the question no. 2

	Year 0	Year 1	Year 2	Year 3	
Total benefits	71,000	35,000	45,000	50,000	2,01,000
Total costs	75,000	71,000	71,000	20,000	2,37,000
Benefit cost	(4,000)	(36,000)	(26,000)	30,000	(36,000)
Cumulative Net cash flow	(4,000)	(40,000)	(66,000)	(36,000)	

So, ROI = $\frac{\text{Total benefit} - \text{total cost}}{\text{total cost}} \times 100\%$

$$= \frac{2,01,000 - 2,37,000}{2,37,000} \times 100\%$$

$$= -15.19\%$$

Break even point = $\frac{\text{number of year of negative cash flow}}{\text{That year's net cash flow} - \frac{\text{That years cumulative cash flow}}{\text{That years net cash flow}}}$

Answer to the question 3

Functional requirement of food delivery:

1. The system will allow user to login or create account using email or contact no.
2. The system will allow users to find nearby restaurant or cafe by giving his/her location.
3. The system will show restaurants based upon the food categories.
4. Users can see the progress of the order.
5. System will allow delivery man to access the location of the user to deliver food and give access of mobile number to contact.

Non-functional requirement of food delivery:

1. Restaurants owner and delivery should not get access of contact and location of a user without ordering their food.
2. App should be bug free.
3. Service provider must verify and provide their information first to register on website.

4. Only administrator should be able to add new service provider.
5. System should get update regular basis to get rid of unwanted bugs.

Ans to the question 4

Functional requirements for ride sharing application:

1. User should be able to set their pickup and drop off point according to their needs.
2. The system ~~also~~ will provide estimated fair according to the road condition and distance.
3. Users and riders will have the freedom to accept or ~~reject~~^{cancel} any order request.
4. The system will provide access to pay fair via cash or online payment method.
5. User should be able to request for only one ride request at a time.

Non-functional requirement of ride sharing application:

1. App should be user friendly so that users can use the app properly.
2. Only admin should have the access of personal information of users and riders.
3. Users can give feedbacks to the riders.
4. ~~The~~ Rider should verify their information before becoming a rider to their app.